

M38813E4HP@QFP64.bin analysis report

Executive identification

- Chip/dump size: Mitsubishi M38813E4HP, MELPS 740-family 8-bit MCU with OTPROM; file size 16,255 bytes / 0x3F7F.
- Likely role: Keyboard controller firmware Version 1.1, RIOS Systems.
- Confidence: High.
- SHA-256: b29761a2fd39abd9c9419ca73b03beb6c41bc52102e2a7429d1f023f82a2a2b8
- MD5: 835fc971bf700ddcc834ef5ba904aaa2
- Byte entropy: 5.1603 bits/byte.

Evidence from this dump

- Offset 0x00001 contains MELPS 740 Series Keyboard Firmware Version 1.1(C) Copyright 1992-1995 RIOS Systems Co., Ltd..
- The contents decode plausibly as Mitsubishi 740-family / 65C02-like MCU code.
- If the image tail is aligned to address 0xFFFF, the inferred base is 0xC081. The probable 6502-style reset vector at 0xFFFFC points to 0xC208, and 0xFFFFE points to 0xE49E.
- Many vectors point to 0xE62C, which looks like a shared default/unused handler area rather than independently meaningful code.

Structural notes

- Active 4 KiB ranges (heuristic, excludes mostly 0x00/0xFF blocks): 0x00000-0x02FFF
- Longest 0xFF runs: 0x25AE len 6575, 0x2388 len 6, 0xDE len 1, 0x234 len 1, 0x30A len 1
- Longest 0x00 runs: 0x22E9 len 76, 0x235D len 14, 0x2374 len 12, 0x236C len 7, 0x217F len 6
- PC option-ROM signatures found:

(none)

Tail vector table, tentative

- Base assumption: image tail maps to 0xFFFF, base = 0xC081.
- 0xFFE0: 0xE62C (inside image, file offset 0x3F5F)
- 0xFFE2: 0xE62C (inside image, file offset 0x3F61)
- 0xFFE4: 0xE62C (inside image, file offset 0x3F63)
- 0xFFE6: 0xE62C (inside image, file offset 0x3F65)
- 0xFFE8: 0xE62C (inside image, file offset 0x3F67)
- 0xFFEA: 0xE62C (inside image, file offset 0x3F69)
- 0xFFEC: 0xE62C (inside image, file offset 0x3F6B)
- 0xFFEE: 0xE62C (inside image, file offset 0x3F6D)
- 0xFFFF0: 0xC0DB (inside image, file offset 0x3F6F)
- 0xFFFF2: 0xE62C (inside image, file offset 0x3F71)
- 0xFFFF4: 0xE62C (inside image, file offset 0x3F73)
- 0xFFFF6: 0xD088 (inside image, file offset 0x3F75)
- 0xFFFF8: 0xE62C (inside image, file offset 0x3F77)
- 0xFFFFA: 0xE62C NMI? (inside image, file offset 0x3F79)

- 0xFFFFC: 0xC208 RESET? (inside image, file offset 0x3F7B)
- 0xFFFFE: 0xE49E IRQ/BRK? (inside image, file offset 0x3F7D)

Interesting strings

- 0x000001: MELPS 740 Series Keyboard Firmware Version 1.1(C) Copyright 1992-1995 RIOS Systems Co., Ltd.

Included supplemental files

- strings/M38813E4HP_keyboard_mcu.strings.txt - all printable strings found by a simple ASCII scanner.
- hexdumps/M38813E4HP_keyboard_mcu_selected_hexdumps.txt - selected annotated hexdumps.
- disassembly/M38813E4HP_keyboard_mcu_reachable_melps740_approx_disasm.txt
- disassembly/M38813E4HP_keyboard_mcu_linear_melps740_approx_disasm.txt

Caveats

- This is static triage, not a full reverse-engineering project. Hardware register names, MCU peripheral vector names, and exact memory maps require board schematics, data books, or emulator instrumentation.
- The MELPS 740 disassembler is approximate. It decodes many 6502/65C02-like opcodes and common bit operations, but Mitsubishi-specific opcodes may be mislabeled or emitted as data bytes.